HW10 / SIGNAL INTEGRITY

Differential channel simulation and equalizer design

Rewritten implementation and numerical verification

A four-port differential channel is simulated with a transmitter feed-forward equalizer (FFE), a continuous-time linear equalizer (CTLE), and a decision-feedback equalizer (DFE). The design sequence follows the assignment: CTLE, then FFE, then DFE. The physical signal path places the FFE before the channel and the other two equalizers at the receiver.

MEETS THE SPECIFIED FINITE-RECORD TARGET

The regenerated post-DFE eye has **144.871 mV** eye height and **15.811 ps** eye width. The assignment requires strict inequalities: EH > 120 mV and EW > 15 ps. These measurements apply to the simulated finite bit record and the model and clock convention stated in this report.

Operating condition	Value
Bit rate / unit interval	40.0 Gb/s / 25.0 ps
Driver rise time	5.0 ps
Each single-ended ideal source swing	2.0 V peak-to-peak
Source and load per conductor	50 ohm / 50 ohm
Simulation record	262,144 bits; seed 0
Time step / samples per UI	0.500 ps / 50

Signal path	Signed EH (mV)	EW (ps)
Channel	-803.741	0.000
Channel + CTLE	-6.779	0.000
FFE + channel + CTLE	32.119	13.736
FFE + channel + CTLE + DFE	144.871	15.811

Negative signed EH means the conditional one/zero populations overlap at the selected clock; the physical eye is closed. The signed value is retained to quantify that overlap.

Channel and voltage normalization

The supplied Touchstone file is a four-port network with a 50-ohm reference impedance. Ports (1, 3) form the input pair and ports (2, 4) form the output pair. The positive conductors are ports 1 and 2. Mixed-mode normalization uses a 100-ohm differential reference and a 25-ohm common-mode reference.

```
Sdd21(f) = [S21(f) + S43(f) - S23(f) - S41(f)] / 2
Hchannel(f) = Sdd21(f) / 2
```

The second factor of one half is the source-to-port impedance division. It is included once in the channel transfer function, so the pulse response, random-bit waveform, and DFE coefficients share the same voltage scale. Sdd21 itself is a wave transmission coefficient; the S-parameter comparison below does not include this source divider.

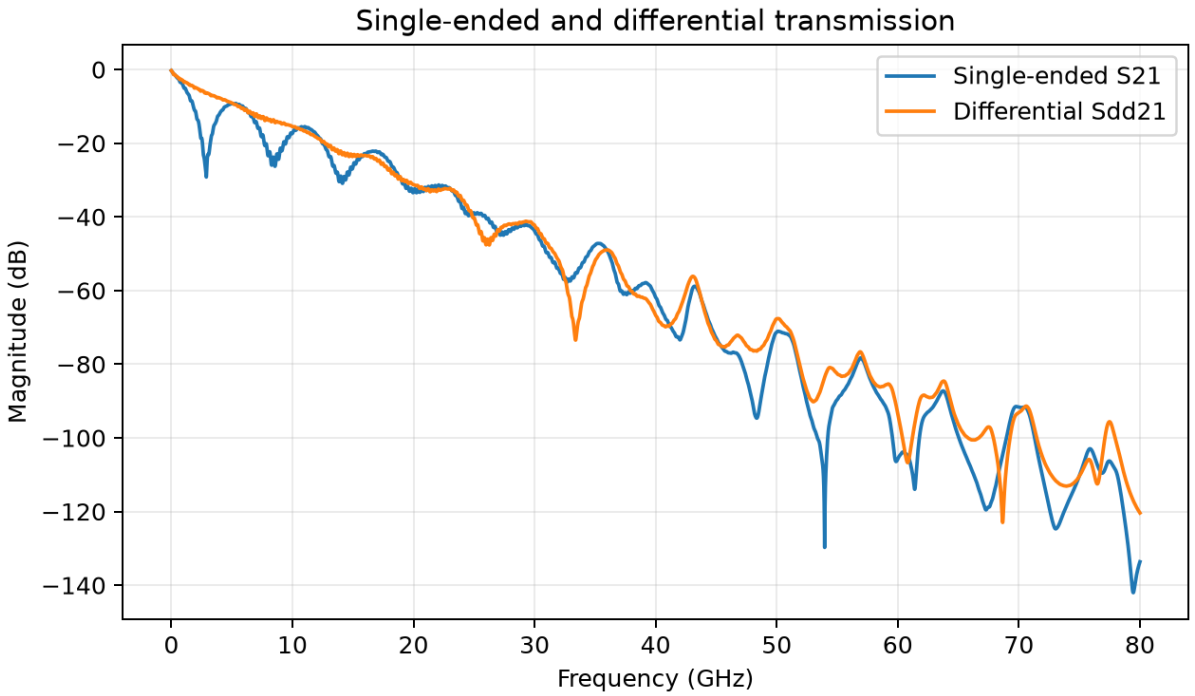


Figure 1. Single-ended S21 and differential Sdd21 of the supplied channel.

Causal transmitter pulse

Data symbols are b[n] in {-1, +1}, with pure differential excitation. The differential source has levels +/-Vpp before loading, where Vpp is the peak-to-peak swing of each single-ended source. A finite-rise single-bit pulse is formed by smoothing a rectangular UI pulse with a causal rise-time rectangle.

```
P(f) = Vpp Tb sinc(f Tb) sinc(f Tr) exp[-j pi f (Tb + Tr)]
SBR(t) = IFT{ P(f) Hchannel(f) W(f) }
y(t) = sum_n b[n] SBR(t - n Tb)
```

Here sinc(x) = sin(pi x)/(pi x). The frequency-domain Kaiser parameter is 1.00. The time-domain response duration is 20.00 ns. Numerically, the causal trapezoid is sampled and convolved with the IDFT channel kernel. The expression above states the corresponding continuous-time pulse convention. Frequency sampling, windowing, truncation, and time discretization are part of the numerical model.

CTLE design and gain limit

The CTLE compensates the high-frequency loss of the differential channel. Its transfer function is expressed in a DC-normalized pole-zero form, with pole and zero frequencies converted to angular frequencies before evaluation.

```
Hctle(s) = Gdc product_z(1 + s/wz) / product_p(1 + s/wp)
Gdc = 10^(DC_gain_dB/20),   wp = 2 pi fp,   wz = 2 pi fz
```

CTLE quantity	Selected value	Assignment limit
Poles (GHz)	40.00000, 40.00000, 40.00000	At most 3; highest <= 40 GHz
Zeros (GHz)	2.52383	No separate zero constraint
DC gain	-9.75161 dB	Included in full gain check
Numerator gain parameter	14.24839 dB	Not the realized peak gain
Maximum gain	5.990000 dB	<= 6 dB
Peak-gain frequency	28.11486 GHz	Numerically located peak

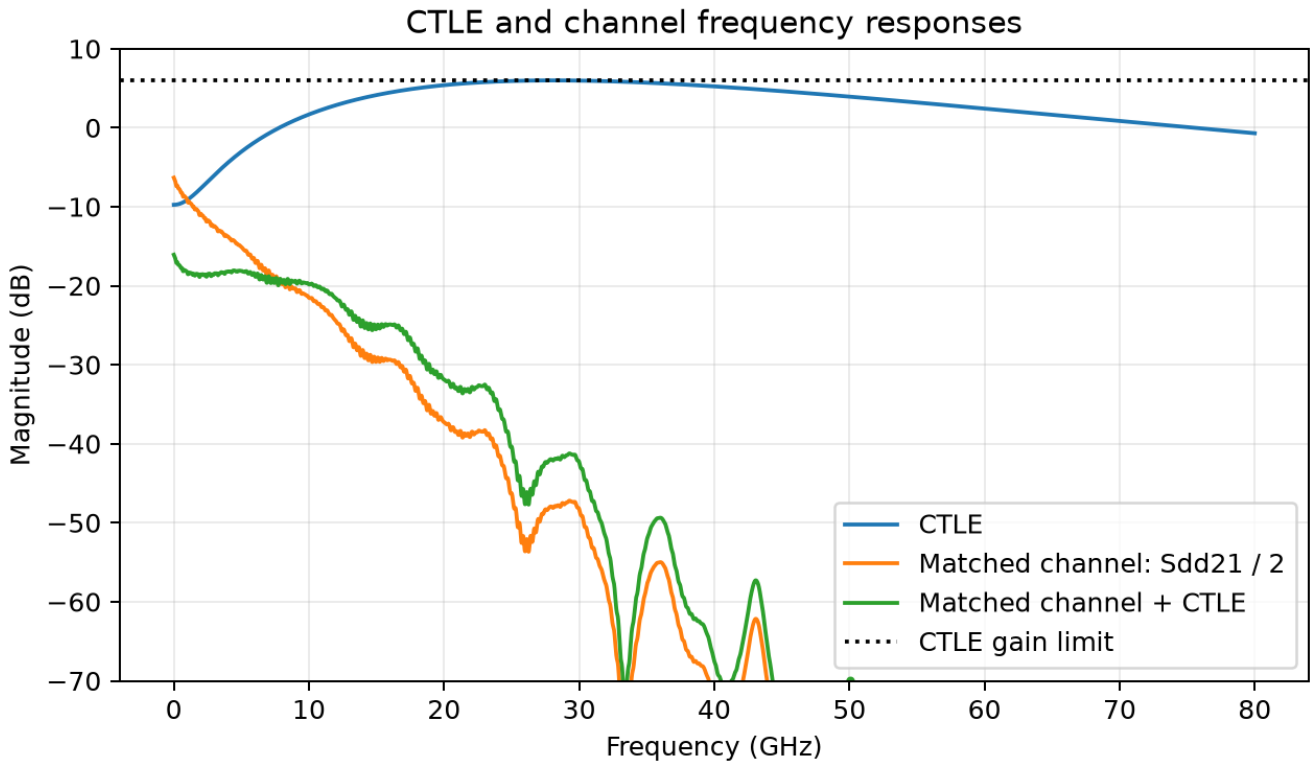


Figure 2. CTLE and combined transfer functions; gain is evaluated over frequency.

A numerator coefficient described as an 'AC gain' in a parameterization does not necessarily equal the maximum magnitude of the realized transfer function. The constraint is checked against the actual peak of |Hctle|, rather than the name or numerical value of a coefficient. The selected equalizer is evaluated together with the channel and transmitter response.

FFE and DFE coefficients

The FFE acts on the differential symbol sequence. Taps are listed in causal order: tap k multiplies b[n-k]. Its absolute coefficient sum fixes the transmitter swing. The DFE subtracts the listed voltage coefficient times the corresponding previous decision; negative coefficients are permitted.

The coefficient search generated 16,384 symbols with seed 70 and retained 14,336 after 1,024-symbol guards at each end. The final settings are checked on separate 262,144-bit records with seeds 0, 1, 2; the validation records are not the coefficient-fitting record.

```
x[n] = sum_k c[k] b[n-k],       sum_k |c[k]| = 1
z[n] = y[n] - sum_i a[i] d[n-i],       d[n] in {-1, +1}
```

FFE tap k	Delay (UI)	Coefficient c[k]
0	0	-0.222313179
1	1	0.777686821

FFE tap count: 2; absolute sum: 1.0000000000. The largest-magnitude coefficient occurs at index 1.

DFE feedback i	Previous decision	Subtracted coefficient (V)
1	d[n-1]	0.039910743
2	d[n-2]	-0.011069638
3	d[n-3]	-0.003789812
4	d[n-4]	0.002306867
5	d[n-5]	0.002296984

DFE threshold: 0.000 V. Decision phase sample: 38 of 50 samples/UI. Each feedback value is held from half a UI before its decision instant up to, but not including, half a UI afterward. The right endpoint uses the next symbol's feedback value. The update convention is identical in waveform synthesis and eye extraction.

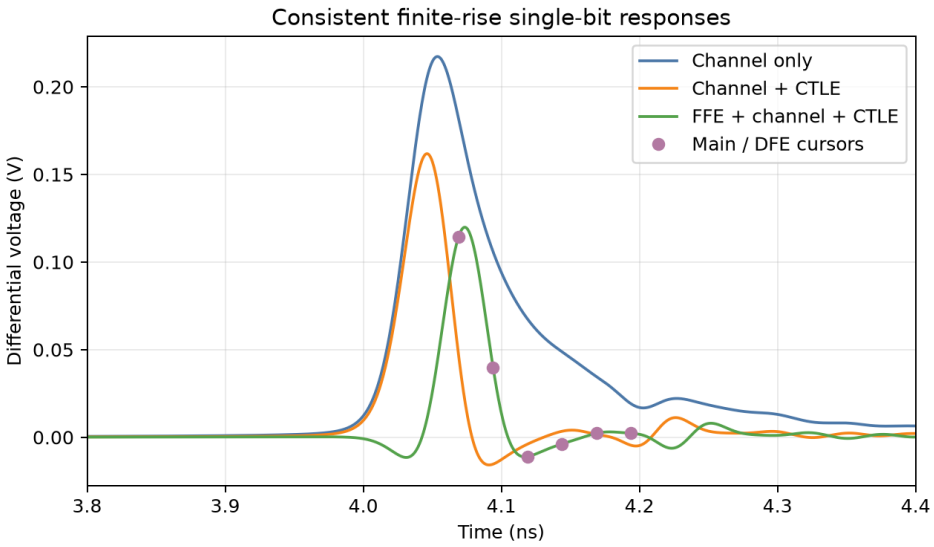


Figure 3. Single-bit responses used to select the linear equalizers and DFE feedback coefficients.

Eye diagrams: channel and CTLE

The same deterministic bit sequence is used across stages. Eye diagrams show the voltage axis in the correct increasing direction. Numerical eye measurements use the underlying trace extrema, rather than reading the image or its histogram bins.

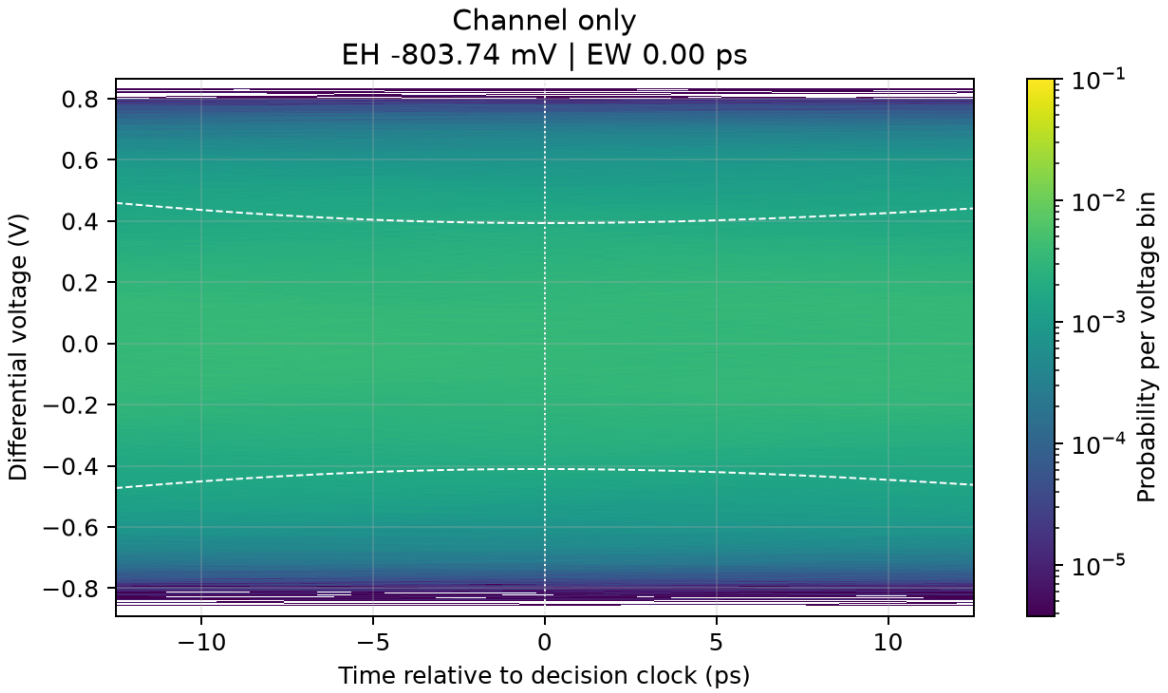


Figure 4. Channel. EH = -803.741 mV; EW = 0.000 ps.

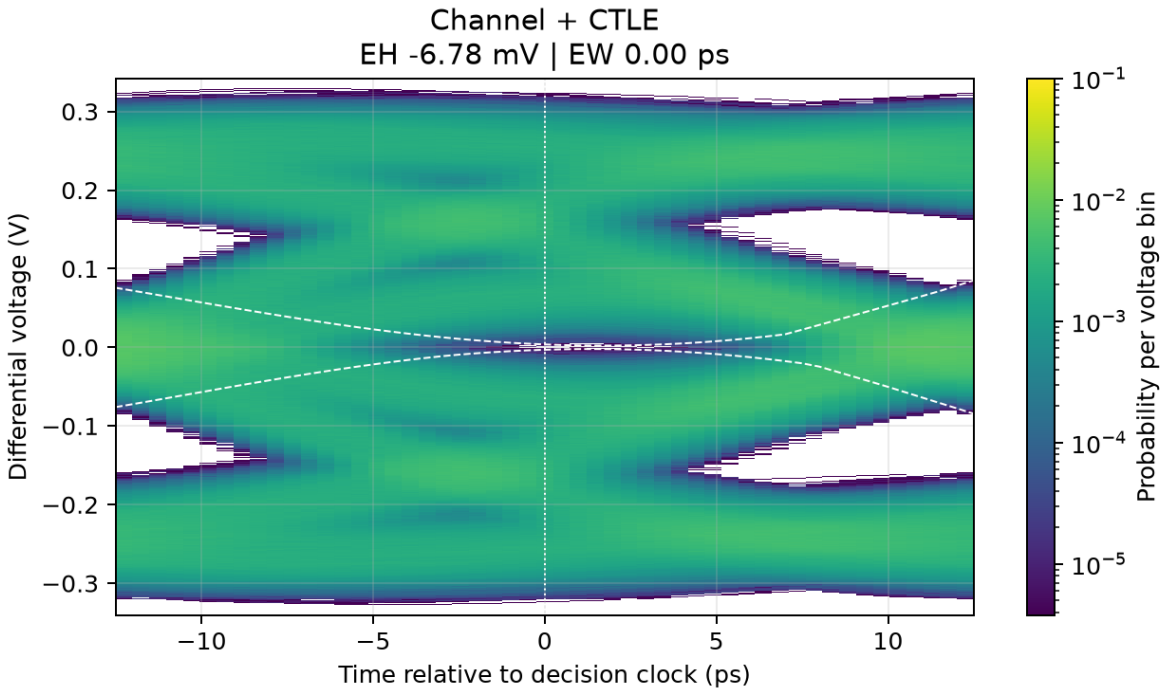


Figure 5. Channel + CTLE. EH = -6.779 mV; EW = 0.000 ps.

Eye diagrams: FFE and DFE

DFE uses its own preceding decisions in the reported simulation. The eye therefore includes any decision-feedback errors that occur in the selected finite record; it is not generated by replacing past decisions with the known transmitted symbols.

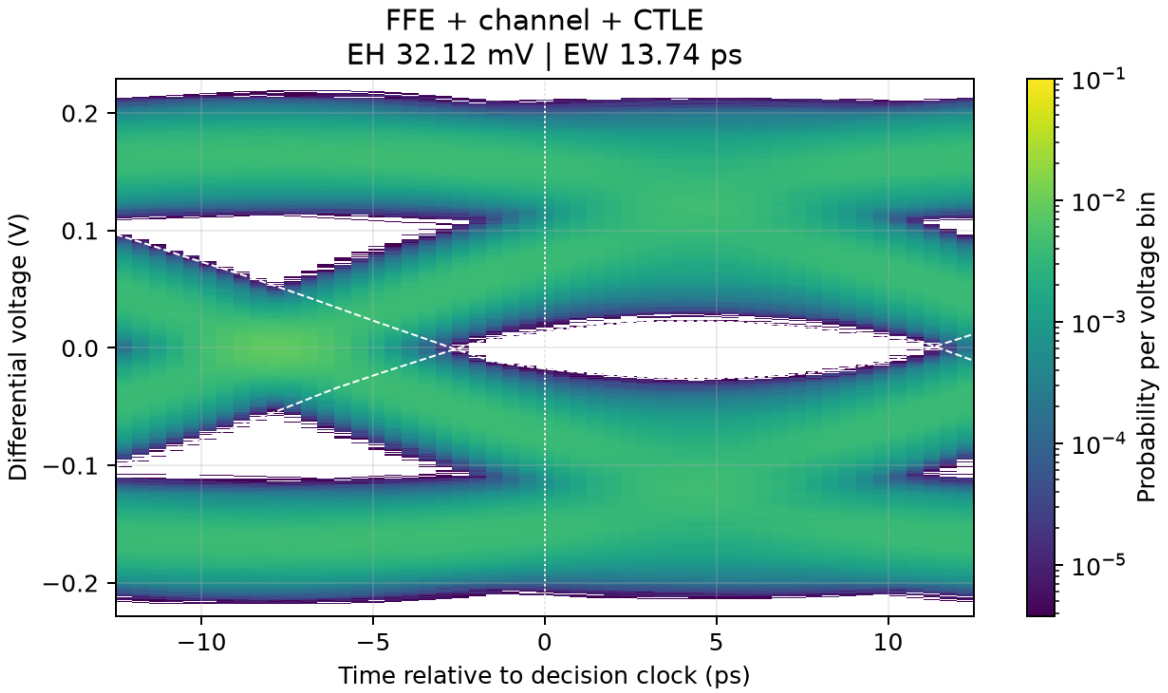


Figure 6. FFE + channel + CTLE. EH = 32.119 mV; EW = 13.736 ps.

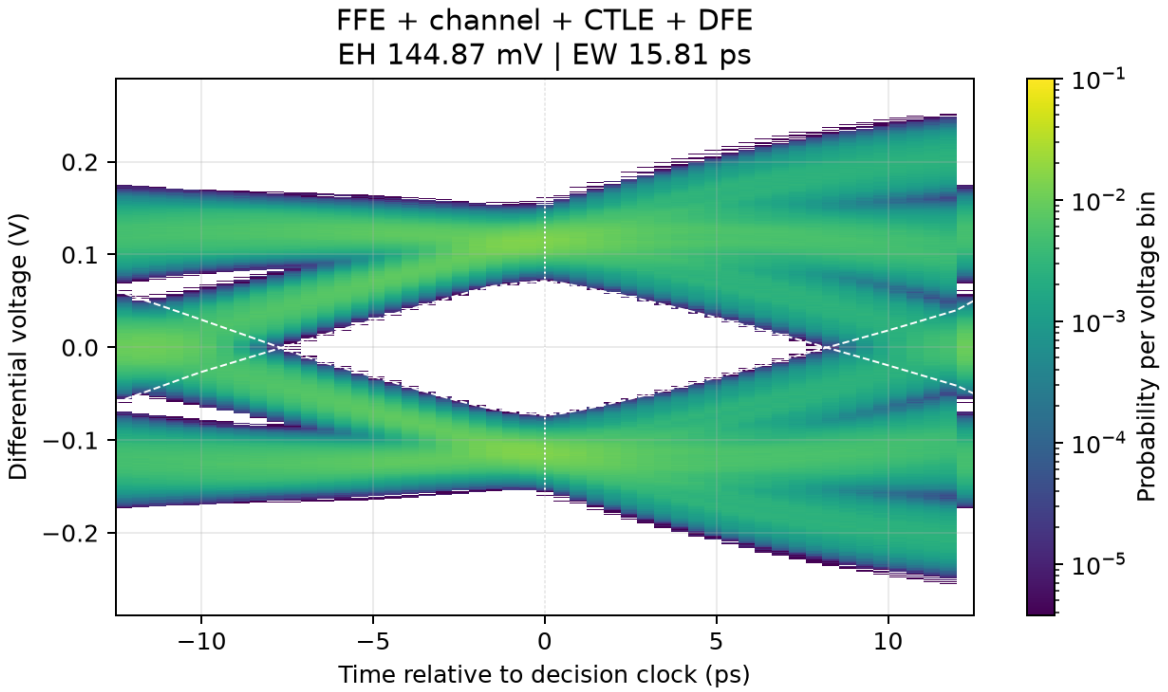


Figure 7. FFE + channel + CTLE + DFE. EH = 144.871 mV; EW = 15.811 ps.

Output waveforms and eye measurement

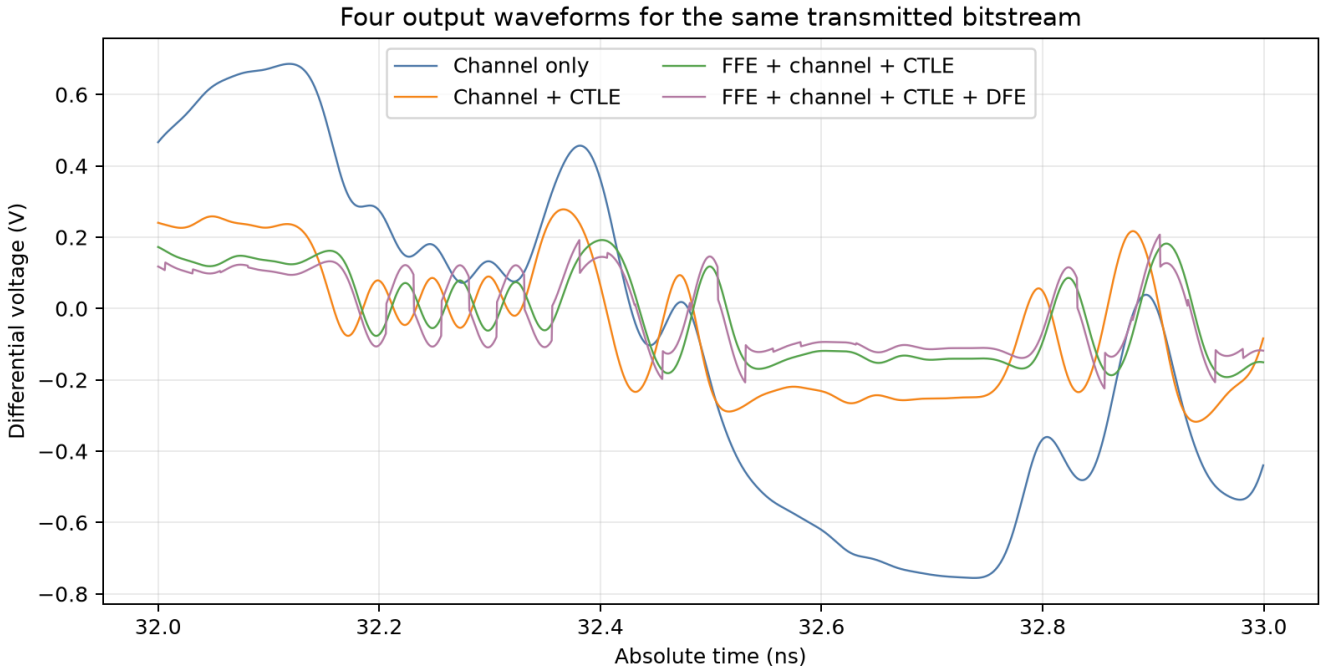


Figure 8. Output waveforms for all four requested signal paths, with their timing convention shown on the plot.

At each phase within one UI, traces are grouped by the transmitted symbol. Eye height is the minimum logic-one voltage minus the maximum logic-zero voltage at the selected decision clock. No percentile clipping is applied. Eye width is the contiguous interval around that clock where every logic-one sample is above 0 V and every logic-zero sample is below 0 V. Its two edges are interpolated using the more restrictive threshold margin.

The analysis retains 262,144 symbols after a 1,024-symbol warmup and adds 1,024 trailing guard symbols. The eye plots are histograms; the reported extrema are computed from every retained trace.

Constraint	Measured or selected value	Limit
Post-DFE eye height	144.871 mV	> 120 mV
Post-DFE eye width	15.811 ps	> 15 ps
CTLE pole count	3	<= 3
Highest CTLE pole	40.00000 GHz	<= 40 GHz
Peak CTLE gain	5.990000 dB	<= 6 dB
FFE / DFE tap counts	2 / 5	<= 5 / <= 5
FFE absolute coefficient sum	1.0000000000	1

The final eye result is a simulation measurement. It is not a statistical BER guarantee and does not certify every possible bit pattern, noise process, or physical implementation.

Verification, corrections, and limits

The legacy waveform applied the source divider after simulation; the rewrite carries it once in the transfer function. It also corrects the dimensionally invalid Tb Tr pulse phase to the causal Tb + Tr convention, uses a common Kaiser setting, aligns the DFE clock, reuses one bit record across stages, and preserves the eye's voltage orientation.

Reproducibility checks

The recorded post-DFE decision-error count is 0. Random seeds, time-step studies, and numerical checks below are retained with the results so that the measurement can be repeated independently.

Seed	Bits	EH (mV)	EW (ps)	Errors
0	262144	144.871	15.811	0
1	262144	141.158	15.967	0
2	262144	143.729	15.881	0

Time step (ps)	Bits	EH (mV)	EW (ps)
0.250	262144	144.852	15.811

Independent verification: **12/12 checks pass**. The separate reconstruction applies FFE to the source levels, then the ramp and time-domain channel convolution, followed by serial DFE and an independently implemented eye measurement. Detailed comparisons are saved in independent_checks.json.

Across all four stages, the largest EH difference is 6.66e-16 V and the largest EW difference is 1.60e-14 ps. The maximum waveform difference over 2,000 checked samples per stage is 9.44e-16 V.

Scope of validation

The model has finite frequency samples and response support, an ideal fixed-rate clock, and idealized equalizers. Finite-record extrema do not guarantee every bit pattern. Noise, jitter, nonlinearity, and adaptation are not modeled. The optional ADS ChannelSim comparison has not been run.

Sources and reproduction

[1] HW10_diff_signaling.pdf, supplied assignment, pp. 1 and 4.
[2] Differential Signaling (1), course handout, pp. 14-17 and 20-31.
[3] Supplied four-port channel and reference implementation. All plotted results are regenerated by the rewritten analysis.

```
python src/hw10.py --validation
python -m unittest discover -s tests -v
python tests/verify_reference.py
python report/build_report.py
```